

GIE — MILK HILL 2001 MATHEMATICAL SUPPORTING RECORD

Formation Geometry, Model Tests, Reproducibility Requirements and Measured-Image Results | Revision 2.0 | 11 August 2026

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1. Formation-Level Exact Structure

The documented formation-level structure is six arms and 409 circles. Its exact arithmetic decomposition is one central circle plus six equal groups of 68 noncentral circles.

$$N = c + k*m = 1 + 6*68 = 409$$

$$\Delta \theta = 2*\pi/6 = \pi/3 = 60 \text{ degrees}$$

This establishes compatibility with one center and six equally populated arms. It does not by itself prove a spiral family, Fibonacci growth law or linguistic message.

2. Ground-Plane Measurement Model

For a full-resolution forensic reconstruction, every circle requires a measured center and radius and aerial perspective must be rectified before raw image coordinates are treated as ground-plane geometry.

$$p_i = [x_i, y_i, 1]^T$$

p'_i proportional to $H p_i$

$$r_i = \sqrt{(x_i - x_0)^2 + (y_i - y_0)^2}$$

$$\theta_i = \text{atan2}(y_i - y_0, x_i - x_0)$$

3. Spiral Model Tests

A spiral model must be selected by measured fit and out-of-sample error, not visual resemblance.

$$\text{Archimedean: } r = a + b*\theta$$

$$\text{Logarithmic: } r = a*e^{(b*\theta)}$$

4. Golden-Ratio / Fibonacci Test

$$\phi = (1 + \sqrt{5})/2 = 1.618033988749895\dots$$

$$r(\theta + \pi/2) / r(\theta) = \phi$$

$$b_{\phi} = 2 \ln(\phi)/\pi = 0.3063489625\dots$$

A golden-spiral claim requires the fitted logarithmic coefficient and its uncertainty to be statistically compatible with b_{ϕ} and outperform reasonable alternatives.

$$t = (b_{\hat{}} - b_{\phi}) / SE(b_{\hat{}})$$

5. Sixfold Rotational Symmetry

A quantitative sixfold test rotates measured geometry in 60-degree increments and compares transformed circles with nearest measured circles while accounting for size.

$$E6 = (1/6N) \sum_j \sum_i \min_l [\text{distance}^2 + \lambda*(s_i - s_l)^2]$$

$E6$ should be compared with $E5$, $E7$ and randomized angular arrangements.

6. Circle-Size Progression

$$\text{Linear: } d_n = \alpha + \beta*n$$

$$\text{Exponential: } \ln d_n = \alpha + \beta*n$$

$$\text{Power law: } \ln d_n = \alpha + \beta*\ln(n)$$

$$\text{Fibonacci residual: } \epsilon_n = d_n - d_{(n-1)} - d_{(n-2)}$$

$$\text{NRMSE}_F = \sqrt{\sum \epsilon_{\text{d}_n}^2 / \sum (\text{d}_n - \text{mean}(\text{d}))^2}$$

The recurrence must be evaluated against measured diameters.

7. Measured-Image Result

The available thumbnail-resolution photograph yielded 361 measurable components. The six-sector phase was 56.921 degrees with sector counts [61, 60, 60, 57, 58, 65]. The fixed diameter-comparison rule produced 355 bits.

$$b_i = 1 \text{ if } d_i > d_{(i-1)}; \text{ otherwise } 0$$

The five-bit sequence did not reproduce the earlier claimed English phrases.

8. Geometry-to-Letter Standard

Any future linguistic decoder must be specified before inspecting text output: fixed geometric start point; traversal direction; feature vector; numerical bins; alphabet table; spacing rule; no manual rearrangement; and blinded testing on another formation.

9. Statistical Test Against Chance

$$P(\text{exact predetermined L-letter string}) = 26^{-L}$$

$$P_{\text{adjusted}} \leq \min(1, M \cdot P_{\text{min}})$$

$$p_{\text{emp}} = [1 + \#\{S_{\text{random}} \geq S_{\text{observed}}\}] / (B + 1)$$

10. Reproducibility Requirement

A complete 409-circle forensic result requires the original full-resolution OH 35mm scan, manual verification of all circles, explicit perspective rectification, publication of the coordinate/diameter table, model comparison, uncertainty estimates and unchanged decoding rules.